Coordenadas Polares

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```
help polar
```

```
  polar Polar coordinate plot.
     polar is not recommended. Use POLARPLOT instead.

     polar(THETA, RHO) makes a plot using polar coordinates of
     the angle THETA, in radians, versus the radius RHO.
     polar(THETA, RHO, S) uses the linestyle specified in string S.
     See PLOT for a description of legal linestyles.

     polar(AX, ...) plots into AX instead of GCA.

     H = polar(...) returns a handle to the plotted object in H.

     Example:
        t = 0 : .01 : 2 * pi;
        polar(t, sin(2 * t) .* cos(2 * t), '--r');

     See also polarplot, plot, loglog, semilogx, semilogy.

     Documentation for polar
```

```
%doc polar
```

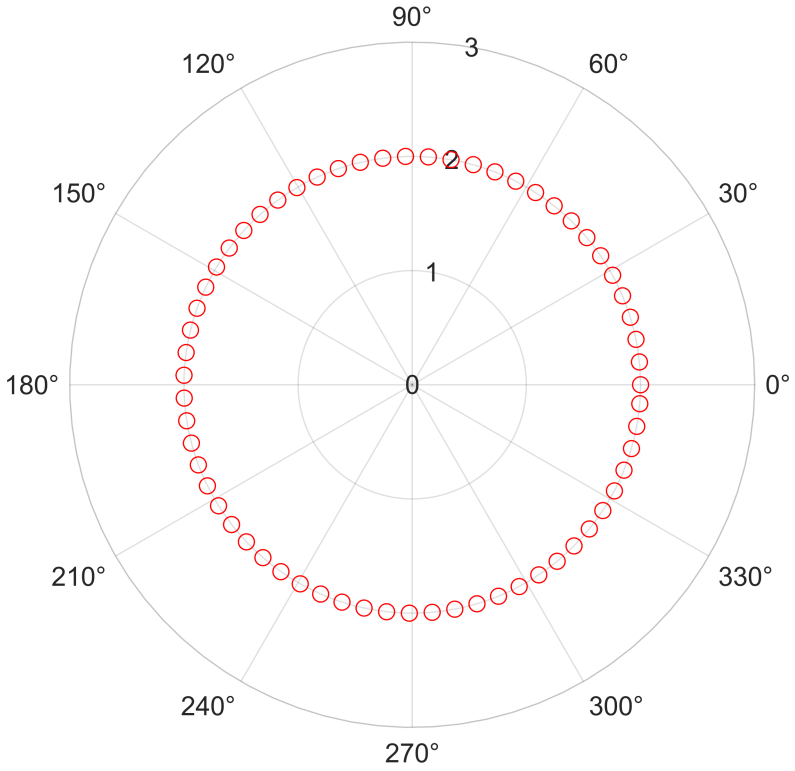
a)

```
[x,y] = Polares2Cartesianas(sqrt(2),pi/4)
```

```
x = 1.0000
y = 1
```

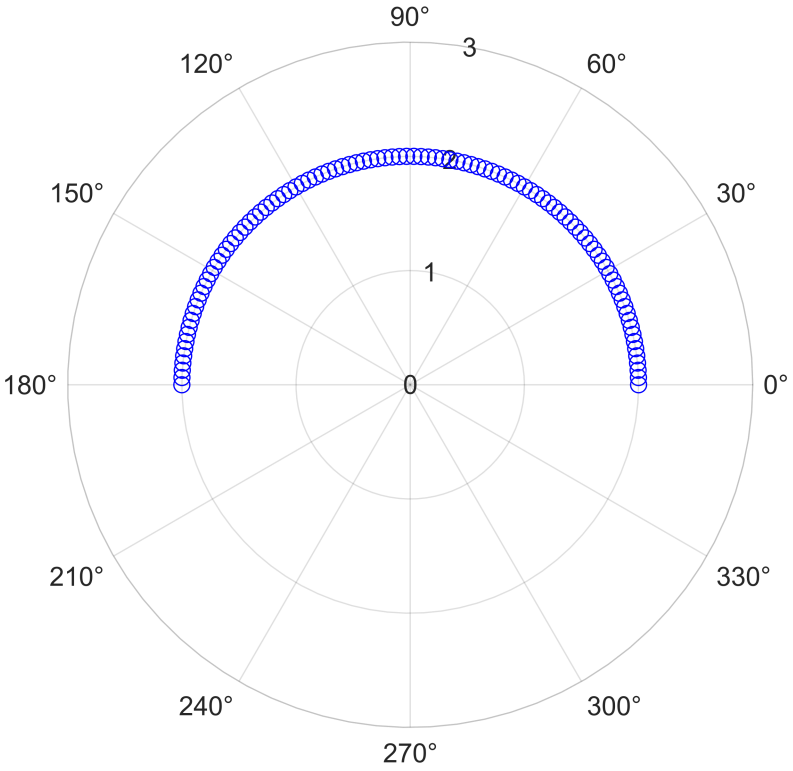
b)

```
theta = 0:0.1:2*pi;
rho = 2;
polarplot(theta,rho,"r-o")
```



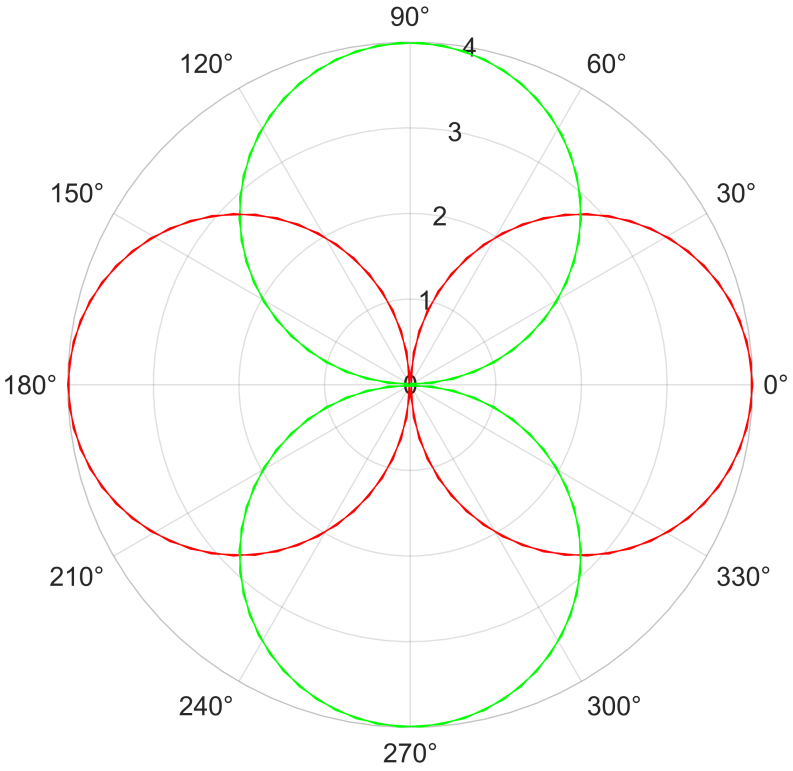
c)

```
theta = linspace(0,pi);
rho = 2;
polarplot(theta,rho,"b-o")
```



d)

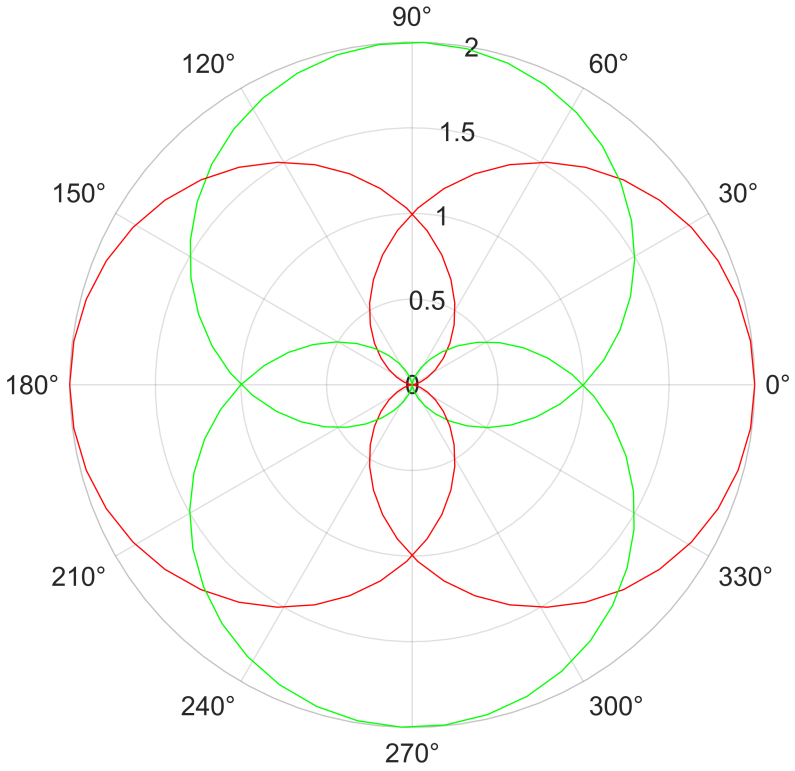
```
theta = linspace(0,2*pi,50);
rho = 4*cos(theta);
polarplot(theta,rho,"r-")
hold on
polarplot(theta,4*sin(theta),"g-")
polarplot(theta,-rho,"r-")
polarplot(theta,-4*sin(theta),"g-")
hold off
```



e) Flor com Pétalas em Cardeoides

```
theta = linspace(0,2*pi,50);
rho1 = 1+cos(theta);
rho2 = 1+sin(theta);

polarplot(theta,rho1,"r-")
hold on
polarplot(theta,rho2,"g-")
polarplot(theta,-rho1,"r-")
polarplot(theta,-rho2,"g-")
hold off
```


90°
120°
60°
150°
30°
180°
0°
210°
330°
240°
300°
270°
2
1.5
1
0.5
0